
Set Theory for Linear Algebra

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Purpose

These notes are intended for undergrads with relatively little proof-writing experience in their first proof-based Linear Algebra course. I'm one year removed from this myself. I recently graded homeworks for this course and found many common gaps in understanding which I myself had also encountered. I hope this document can address some of these, in particular those regarding the fundamental constructions of sets, and how to write proofs with them in a Linear Algebra context. I include many examples for concepts I imagine are new to most, less examples for more familiar concepts, and some exercises. All mistakes are my own.

I do not claim expertise over anything I will discuss here. In fact, my knowledge is completely limited to undergrad level algebra. I hope this will be somewhat of an asset in writing these notes, as I can very closely relate to struggling with these concepts. As such, this document is essentially from the perspective of a peer tutor.

Most examples assume prerequisite knowledge up to (inclusively) the usual treatment of abstract vector spaces, where these concepts become of paramount importance. However, outside specific examples, the rest of the notes are accessible given just a couple Linear Algebra lectures. I redefine some objects of interest as reinforcement.

For the most part, I introduce rigor when it is beneficial for intuition. I do not attempt to go far beyond the scope of what is needed for introductory Linear Algebra. I present tautologies or very intuitive properties as “Facts” and avoid proving them exhaustively; belaboring these can be extremely useful, but would constitute filler for this document. I include exercises, some borrowed from Linear Algebra: A Geometric Approach by Shifrin and Adams [SnA].

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1 Sets

1.1 Basics

I imagine most prospective Math majors have seen and worked with sets to some extent before. For instance, those with Multivariable Calculus background may recognize the set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ as the unit circle. I hope to make some bits more rigorous. Here, I borrow heavily from Prof. Terence Tao's *Analysis I* textbook [Tao], in particular Chapter 3 on Set Theory. Read this for a properly complete treatment.

Definition 1.1.1. (*Sets*). A *set* S is an unordered collection of *objects* which we call the *elements* of S . If the object a is an element of S , we write $a \in S$, or “ a in S .” We typically use the curly brackets $\{\cdot\}$ to define sets.

Example 1.1.2. (i) $S = \{1, 5, 2, 9\}$ is a collection of four numbers, (ii) $S = \mathbb{R}$ is the set of real numbers, (iii) $S = \emptyset$, the empty set (see Axiom 1.3).

What is an object? For our purposes: some *thing* of mathematical interest. Rigorously defining an object may be more of a philosophical issue. For now, we only need to worry about the typical objects in Linear Algebra:

- Natural numbers, which make up the set $\mathbb{N}$. Tao defines these semi-rigorously in Chapter 2 of [Tao]. Informally, these are just numbers we use to count distinct physical things. We will take 0 to be in $\mathbb{N}$. Note infinity is not in $\mathbb{N}$.
- Integers, which make up the set $\mathbb{Z} := \{m - n : m, n \in \mathbb{N}\}$.
- Rationals, which make up the set $\mathbb{Q} := \{m/n : m, n \in \mathbb{Z}, n \neq 0\}$
- Real numbers, which make up the set $\mathbb{R}$. How $\mathbb{R}$ is constructed is a key topic in a first Real Analysis course. For now, think of $\mathbb{R}$ as the rationals with all the gaps (e.g. $\sqrt{2}$, π , e) filled in.
- Vectors in $\mathbb{R}^n$ (or any vector space).
- Functions, typically linear transformations (and by extension, matrices) in the Linear Algebra context.
- Sets themselves (and by extension, vector spaces, subspaces, bases, etc.). This is established axiomatically (see Axiom 1.1).

Remark 1.1.3. Here, we use what is sometimes called *set-builder notation* to define $\mathbb{Z}$ and $\mathbb{Q}$, which looks like $S = \{P(x) : Q(x)\}$, where $P(x)$ and $Q(x)$ are some expressions involving an object x . Read this as “ $P(x)$ such that $Q(x)$.” The validity of this notation can be justified axiomatically (see Axioms 1.4, 1.5). I will also use the symbol “ $:=$ ” to mean “defined as.”

Definition 1.1.4. (*Subset*). A set S' is a *subset* of a set S , denoted $S' \subseteq S$, if and only if every element of S' is an element of S . In this scenario, we call S the *superset* of S' .

Example 1.1.5. (i) $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$. (ii) $\{5, 6\} \subseteq \{1, 5, 6, 8\}$. (iii) For $S = \{1, 2, \{3, 4\}\}$, we have $\{1, 2\} \subseteq S$ but $\{1, 2\} \notin S$, and $\{3, 4\} \not\subseteq S$ but $\{3, 4\} \in S$. This is because the set $\{3, 4\}$ is an element of S and not a subset, as $3 \notin S$ and $4 \notin S$.

Fact 1.1.6. *Let A, B, C be sets. Then the following hold:*

- *If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.*
- *If $A \subseteq B$ and $B \subseteq A$ then $A = B$ (we prove this in Theorem 3.1.2).*

1.2 Axioms

Here I will state the important axioms of set theory for our purposes, borrowed again from Tao [Tao]. This list is incomplete with respect to a full ZFC axiomatic model of set theory (which underpins the logical validity of most things), but contains what, to me, are the most beneficial for intuition in proof-writing. Thus, there will technically be some logical holes, which we will not belabor. Read [Tao] or ask an LLM for more details.

Definition 1.2.1. (*Axiom*; informal). In math, an *axiom* is a statement taken to be true without the need for justification. We state axioms as fundamental assumptions, which serve to define the logical framework for building further arguments.

Remark 1.2.2. Note that axioms differ from theorems, propositions, lemmas and corollaries, all of which need proof to validate their logical correctness.

Axiom 1.1. (*Sets are objects*). If S is a set, then S is an object.

Axiom 1.2. (*Equality of sets*). Two sets A and B are *equal* if and only if every element of A is an element of B , and every element of B is an element of A . Symbolically:

$$A = B \iff y \in A \forall y \in B \text{ and } x \in B \forall x \in A$$

Remark 1.2.3. Intuitively, set equality holds only when two sets contain the exact same elements. Note that sets are defined as unordered collections, so equality is also order independent, where by order I mean which element is listed first, not some notion of magnitude of elements, i.e. $\geq$ or $\leq$.

Axiom 1.3. (*Empty set*). There is an *empty set*, denoted $\emptyset$, which contains no elements. In other words, for any object a , it holds that $a \notin \emptyset$. We call all sets with some element, i.e. all sets S for which there exists an object $a \in S$, *nonempty*.

The following two axioms set up the notion of building sets out of objects which satisfy some property, which is done symbolically with set-builder notation.

Definition 1.2.4. (*Property*; informal). Let x be an object. A *property* of x , usually denoted $P(x)$, is a statement involving x that is either true or false once x is specified. For a specified object x , if $P(x)$ is true, we say x *satisfies* $P(x)$, or that $P(x)$ *holds*.

Axiom 1.4. (*Axiom of Specification*). Let S be a set. For each element $x \in S$, let $P(x)$ be a property of x . Then there exists a set comprised of precisely the elements $x \in S$ for which $P(x)$ holds. We denote this set $\{x \in S : P(x)\}$.

Remark 1.2.5. Axiom 1.4 permits us to define something like $\{x \in \mathbb{R} : x > 0\}$, where $S = \mathbb{R}$ and $P(x) = (x > 0)$. In other words, we can now define sets whose elements satisfy a given property, as subsets of “bigger” sets. This essentially sets up one half of set-builder notation.

Example 1.2.6. Some familiar examples leveraging the Axiom of Specification:

- $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ the unit circle. Here $S = \mathbb{R}^2$ and $P(x, y) = (x^2 + y^2 = 1)$. Note that $P(\cdot)$ becomes an expression of two variables due to the structure of $\mathbb{R}^2$ (discussed in Section 2.4).
- $\{x \in \mathbb{R}^n : a \cdot x = 0\}$ for some vector $a \in \mathbb{R}^n$, which defines a hyperplane. Here $S = \mathbb{R}^n$ and $P(x) = (a \cdot x = 0)$.
- $\{x \in \mathbb{R}^n : Ax = 0\}$ for some $m \times n$ real matrix A , which defines the null space or kernel of A . Here $S = \mathbb{R}^n$ and $P(x) = (Ax = 0)$.
- $\{y \in \mathbb{R}^m : \exists x \in \mathbb{R}^n \text{ where } y = Ax\}$ for some $m \times n$ real matrix A , which defines the column space or image of A . Here $S = \mathbb{R}^m$ and $P(y) = (\exists x \in \mathbb{R}^n \text{ where } y = Ax)$. Note the variables are renamed to match Linear Algebra convention.

Axiom 1.5. (*Axiom of Replacement*). Let S be a set. For each element $x \in S$ and arbitrary object z , let $P(x, z)$ be a property of x and z such that for each $x \in S$ there exists at most one object z for which $P(x, z)$ is true. Then there exists a set comprised of precisely the objects z for which $P(x, z)$ holds for some $x \in S$. We denote this set $\{z : P(x, z) \text{ for some } x \in S\}$. If $P(x, z) = (z = Q(x))$ for some property $Q(x)$ of x , we abbreviate the notation to $\{Q(x) : x \in S\}$.

Remark 1.2.7. Axiom 1.5 permits us to define something like $\mathbb{C} := \{a + bi : a, b \in \mathbb{R}\}$ the complex numbers (where i denotes the imaginary unit satisfying $i^2 = -1$), where $S = \mathbb{R}$ and $P(a, b, z) = (z = a + bi)$. In other words, we can now define sets as, in a sense, “functions” of other sets, which may be “smaller” in a non-rigorous sense. This gives the other half of set-builder notation.

Example 1.2.8. Some familiar examples leveraging the Axiom of Replacement:

- $\{x^2 : x \in \mathbb{Z}\}$ the set of squared integers. Here $S = \mathbb{Z}$ and $P(x, z) = (z = x^2)$.
- $\{m - n : m, n \in \mathbb{N}\}$ and $\{m/n : m, n \in \mathbb{Z}, n \neq 0\}$, giving $\mathbb{Z}$ and $\mathbb{Q}$ respectively.
- $\{z : z = 0 \text{ for some } x \in \mathbb{R}\}$. This looks odd at first, but is really just the singleton set $\{0\}$, as $S = \mathbb{R}$ and $P(x, z) = (z = 0)$, so $P(x, z)$ only holds for the object 0, independent of the value of x .

- $\{Ax : x \in \mathbb{R}^n\}$ for some $m \times n$ real matrix A , which defines the column space or image of A . Here $S = \mathbb{R}^n$ and $P(x, z) = (z = Ax)$. Note this is equivalent to the definition in Example 1.2.6.

Example 1.2.9. We can combine the Axioms of Specification and Replacement in set definitions. For instance: $\{1/x : x \in \mathbb{R}, x \neq 0\}$. The definition of $\mathbb{Q}$ given in Example 1.5 also uses both axioms.

Remark 1.2.10. I choose to leave out a discussion of the Axiom of Choice, though the machinery in the subsequent Sections 1.3-2.4 is sufficient to make an informal statement. See Chapter 8 in [Tao] or any online resource for more.

1.3 Choosing Elements from Sets

Here I discuss the notion of picking elements from sets, which is a natural step in set-related proofs but can be an unobvious weapon initially to many students (including myself). A key goal is to be able to take certain elements from sets and use their properties to argue about the set as a whole.

Fact 1.3.1. (Single Choice). *If S is a nonempty set, then there exists an element in S . Hence we can pick an arbitrary element $x \in S$, and work with this element in subsequent arguments.*

Remark 1.3.2. Tao proves Fact 1.3.1 as a Lemma following Axioms 1.2 and 1.3 (all that is needed is the existence of some element) in [Tao], as it is needed to formally establish the Axioms of Specification and Replacement. We will take it as a proof strategy: take an arbitrary element from a set and use its properties to prove something general about the set itself. Note this logic is valid, as we picked an element with no “special” properties, so any argument pertaining to this element applies to any element in the set.

Example 1.3.3. (i) Let $S = \mathbb{N}$, which is nonempty. If we take an arbitrary $x \in S$, we know, for instance, that $x \geq 0$ from basic arithmetic (technically following Peano’s Axioms for natural numbers). (ii) Let $S = \{x \in \mathbb{R}^n : Ax = 0\}$ for some $m \times n$ real matrix A , which is nonempty because $0 \in S$. If we take an arbitrary $x \in S$, we know that $Ax = 0$.

Again, note that the ability to pick an arbitrary element from a nonempty set seems trivial, but is extremely powerful in proof-writing for essentially every proof-based course (see examples in Section 3). Next, we will validate the act of picking “specific” or “distinguished” elements from a set, instead of a completely arbitrary one.

Fact 1.3.4. (Distinguished Choice). *Let S be a nonempty set. For each element $x \in S$, let $P(x)$ be a property of x . If there exists some element $x \in S$ such that $P(x)$ holds, then we can pick such an element and work with it in subsequent arguments.*

Remark 1.3.5. The assumption that some element $x \in S$ satisfies $P(x)$ is equivalent to assuming the set $\{x \in S : P(x)\}$, which exists by the Axiom of Specification, is nonempty. We now

have another proof strategy: if we know some element with some property exists in a set, we can take such an element and argue about the set.

Remark 1.3.6. Note that we are not using a new principle beyond Single Choice in invoking Distinguished Choice; we are still just using the nonemptiness of a set. I choose to separate these, as they are used in distinct manners in most Linear Algebra proofs.

Example 1.3.7. (i) Let $S = \mathbb{Z}$, which is nonempty and contains 2, meaning there exists some element $x \in S$ for which $P(x) = (x = 2)$ holds, namely 2 itself. Then we can take $2 \in \mathbb{Z}$. (ii) Let $S = \mathbb{R}^n$, which is nonempty, and let $\{v_1, \dots, v_n\}$ be a set of linearly dependent vectors in $\mathbb{R}^n$. Then we know there exists some coordinates $(c_1, \dots, c_n) \in \mathbb{R}^n$, which is nonzero and satisfies $P(c_1, \dots, c_n) = (\sum_{i=1}^n c_i v_i = \mathbf{0})$. Thus we can pick $(c_1, \dots, c_n) \in \mathbb{R}^n$, where $c_1, \dots, c_n$ are not all zero, such that $\sum_{i=1}^n c_i v_i = \mathbf{0}$.

Distinguished Choice can be particularly useful for disproving some structural property of certain sets. For these problems, it suffices to exhibit a specific element (or collection of specific elements) that leads to a contradiction (see Section 3 for examples).

Lastly, consider the act of choosing elements from multiple sets simultaneously. For a finite number of nonempty sets, we can extend Facts 1.3.1 and 1.3.4 to a Cartesian product of indexed sets (see Sections 2.2 and 2.4 for what these mean); this amounts to the fact that Cartesian products of nonempty sets are nonempty.

However, for an infinite number of nonempty sets, we need to invoke the Axiom of Choice to pick elements. An exception occurs in a special case of Distinguished Choice, where the property $P(x)$ uniquely specifies an element in each set in the collection, i.e. gives a *uniform choice-free selection rule*. In other words, we need “there exists a unique” element satisfying $P(x)$, not just “there exists” an element satisfying $P(x)$. Note that we need the rule to be uniform across every set in the collection; we can never change the rule. In this case, we are not really making a choice; we are picking one specific element from each set, and we know exactly which element it is for every set.

Example 1.3.8. Some simple examples involving a uniform choice-free selection rule:

- Consider the collection of all finite-dimensional vector spaces. By the vector space axioms, we know each must contain the $\mathbf{0}$ vector as an additive identity, which is unique. Then we can select $\mathbf{0}$ from each vector space without invoking choice.
- Consider the singleton collection consisting of $S = \mathbb{R}^n$. Invoke Single Choice to pick $b \in \mathbb{R}^n$. Then by the uniqueness of additive inverses, we know there exists a unique $x \in S$ satisfying $P(x) = (x + b = \mathbf{0})$, namely $-b$. Thus we can select $-b \in \mathbb{R}^n$ without invoking choice.

If we do not have a uniform choice-free selection rule, then choosing arbitrary elements, or elements that satisfy a property that may pertain to multiple elements, requires the Axiom of

Choice, which essentially validates this act. Stating the axiom requires more formal definitions of what *choice* means. Luckily, we usually don't worry about this in Linear Algebra.

2 More to do with Sets

Now that we have some foundational structure for sets, we can discuss more “second-order” (not in the logical sense, but in the “how we can think about them” sense) concepts that build upon Section 1.

2.1 Functions on Sets

We discuss here the notion of defining functions on sets. A familiar trait of functions from high school math is that they pass the “vertical line test”; this can be made rigorous. I will give fewer examples, as I find most students are much more comfortable working with functions than sets.

Definition 2.1.1. (*Function*; informal). Let S and S' be sets. A *function* f (or *mapping*) from S to S' , denoted $f : S \rightarrow S'$, is an operation which assigns exactly one element $x' \in S'$ to each element $x \in S$. We denote the element in S' assigned by f to some *input* element $x \in S$ as the *output* element $f(x)$.

Definition 2.1.2. (*Function*). Let S and S' be sets, and let $P(x, x')$ be a property of objects $x \in S$ and $x' \in S'$. Suppose for every $x \in S$, there is exactly one $x' \in S'$ for which $P(x, x')$ holds. Then we can use $P(x, x')$ to define a *function* $f : S \rightarrow S'$ which, for any input $x \in S$, assigns the unique object $f(x) \in S'$ for which $P(x, f(x))$ holds as the output. We call S the *domain* of f and S' the *codomain* of f .

Remark 2.1.3. Both Definitions 2.1.1 and 2.1.2 have the vertical line test notion baked in, as both assert that functions must assign exactly one output for each input. Additionally, note that the domain and codomain are intrinsic properties of functions.

Remark 2.1.4. We often define a function $f : S \rightarrow S'$ by specifying what it does to an arbitrary $x \in S$. In particular, we declare functions by stating “let f be the function $S \rightarrow S'$ given by $x \mapsto f(x)$ for all $x \in S$,” where we explicitly write the expression for $f(x)$ with respect to x .

Example 2.1.5. (i) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto x^2$ for all $x \in \mathbb{R}$ is a function. (ii) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto 1/x$ is not technically a function, as $f(0) \notin \mathbb{R}$. However, $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ is a function, where $\mathbb{R} \setminus \{0\}$ denote all nonzero real numbers. (iii) Let A be an $m \times n$ real matrix. Then $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $x \mapsto Ax$ for all $x \in \mathbb{R}^n$ is a function. (iv) Let $C^1(\mathbb{R})$ be the set of once differentiable functions $g : \mathbb{R} \rightarrow \mathbb{R}$, and let $C^0(\mathbb{R})$ be the set of continuous functions $g : \mathbb{R} \rightarrow \mathbb{R}$. Then $f : C^1(\mathbb{R}) \rightarrow C^0(\mathbb{R})$ given by $g \mapsto g'$ for all $g \in C^1(\mathbb{R})$, where g' denotes the first derivative of g , is a function.

Definition 2.1.6. (*Function equality*). Let $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ be functions. We say f and g are *equal* if $X = Y$, $X' = Y'$, and $f(x) = g(x)$ for all $x \in X$. Note that we require the domains and codomains to be equal setwise.

Definition 2.1.7. (*Function Composition*). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. We define the *composition* of f and g , denoted $g \circ f$, as the function $A \rightarrow C$ given by $x \mapsto g(f(x))$ for all $x \in A$. In other words, $(g \circ f)(x) := g(f(x))$.

Definition 2.1.8. (*Identity*). The function $S \rightarrow S$ given by $x \mapsto x$ is called the *identity* function on S , sometimes denoted id_S . More formally, we define the identity as the (unique) function $\text{id}_S : S \rightarrow S$ satisfying

$$f \circ \text{id}_S = f \text{ and } \text{id}_{S'} \circ f = f$$

for all functions $f : S \rightarrow S'$.

Definition 2.1.9. (*Injection*). A function $f : S \rightarrow S'$ is called *injective*, or an *injection*, if it maps different inputs to different outputs. Formally: f is injective if for all $x, y \in S$, $x \neq y \implies f(x) \neq f(y)$. Equivalently, we can use the contrapositive: f is injective if $f(x) = f(y) \implies x = y$ for all $x, y \in S$. We also call injections *one-to-one*.

Definition 2.1.10. (*Surjection*). A function $f : S \rightarrow S'$ is called *surjective*, or a *surjection*, if every element in S' is the output for some input in S . Formally: f is surjective if for every $x' \in S'$, there exists $x \in S$ for which $x' = f(x)$. We also call surjections *onto*.

Definition 2.1.11. (*Bijection*). A function $f : S \rightarrow S'$ is called *bijective*, or a *bijection*, if f is both injective and surjective. We also call bijections *one-to-one correspondences*.

Remark 2.1.12. Note that whether a function is injective, surjective, or bijective depends on its domain and codomain. For instance, $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto |x|$ is not surjective, but $\tilde{f} : \mathbb{R} \rightarrow [0, \infty)$ given by $x \mapsto |x|$ is surjective. Similarly, $g : \mathbb{Z} \rightarrow \mathbb{Z}$ given by $x \mapsto x^2$ is not injective, but $\tilde{g} : \mathbb{N} \rightarrow \mathbb{Z}$ given by $x \mapsto x^2$ is injective.

Definition 2.1.13. (*Permutation*). A bijection $\sigma : S \rightarrow S$, i.e. a bijection where the domain and codomain are equal setwise, is called a *permutation* of S .

Definition 2.1.14. (*Inverse*). Let $f : S \rightarrow S'$ be a function. A function $g : S' \rightarrow S$ is called the *inverse* of f if $f \circ g = \text{id}_{S'}$ and $g \circ f = \text{id}_S$. We denote the inverse of f as f^{-1} . We call a function f *invertible* if it has an inverse.

Definition 2.1.15. (*Image of subsets*). Let $f : S \rightarrow S'$ be a function, and let $A \subseteq S$. We call the set of outputs in S' given by elements in A under f , denoted $f(A)$, the *image* of A under f . Formally:

$$f(A) := \{x' \in S' : x \in A\} \subseteq S'$$

Definition 2.1.16. (*Image of functions*). Let $f : S \rightarrow S'$ be a function. We call the set of outputs in S' given by all elements in S the *image* of f , denoted $\text{image}(f)$. Formally:

$$\text{image}(f) := f(S) = \{x' \in S' : x \in S\} \subseteq S'$$

Remark 2.1.17. The image of a function $f : S \rightarrow S'$ is just the image of subsets specialized to when the subset is the whole domain S .

Definition 2.1.18. (*Preimage of subsets*) Let $f : S \rightarrow S'$ be a function, and let $A' \subseteq S'$. We call the set of inputs in S mapping to elements in A' under f , denoted $f^{-1}(A')$, the *preimage* of A' under f . Formally:

$$f^{-1}(A') := \{x \in S : f(x) \in A'\} \subseteq S$$

Remark 2.1.19. Note that we do not require f to be invertible for preimages of subsets to be well-defined. The “ f^{-1} ” in the preimage notation is not the inverse of the function f ; it is just convenient notation. A common special case is when f is invertible and A' is specialized to a single element x' . Then the preimage of x' is $x := f^{-1}(x') \in S$, where f^{-1} now denotes the inverse of f .

Fact 2.1.20. Let $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$ be functions. With basic definitions above, the following properties hold:

- (i) (*Associativity of composition*). $(h \circ g) \circ f = h \circ (g \circ f)$.
- (ii) If f and g are injective, then $g \circ f$ is injective. If f and g are surjective, then $g \circ f$ is surjective. If f and g are bijective, then $g \circ f$ is bijective.
- (iii) If f and g are invertible, then $g \circ f$ is invertible. In particular, the inverse of $g \circ f$ is given by $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- (iv) f is bijective if and only if f is invertible (we prove this in Theorem 3.4.2).
- (v) f is surjective if and only if $\text{image}(f) = B$.
- (vi) It holds that $\text{image}(g \circ f) = g(\text{image}(f))$.

2.2 Indexing Sets

Here I discuss the notion of indexing a set, which requires some more machinery involving cardinalities.

Definition 2.2.1. (*Cardinality*; informal). The *cardinality* of a set S is an intrinsic notion of size, denoted $|S|$ or $\text{card } S$, which roughly describes the number of elements in S . Sets S and S' have *equal cardinality* if and only if there exists a bijective map $f : S \rightarrow S'$.

Fact 2.2.2. Let X, Y, Z be sets. Some properties of cardinality include:

- (i) (*Reflexive*) X and X have equal cardinality.
- (ii) (*Symmetric*) If X and Y have equal cardinality, then Y and X have equal cardinality.
- (iii) (*Transitive*) If X and Y have equal cardinality, and Y and Z have equal cardinality, then X and Z have equal cardinality.

Definition 2.2.3. (*Finite cardinality*). A set S has cardinality $n \in \mathbb{N}$ if and only if it has equal cardinality with the set $\{i \in \mathbb{N} : 1 \leq i \leq n\}$, i.e. S has n elements. If S has cardinality $n \in \mathbb{N}$, we call S *finite*. Otherwise, we call S *infinite*.

Example 2.2.4. The sets $S = \{5, 7, 9\}$ and $S' = \{0.5, \pi, -1\}$ have equal cardinalities of 3, as we can explicitly construct bijective maps that associate $5 \leftrightarrow 0.5 \leftrightarrow 1$, $7 \leftrightarrow \pi \leftrightarrow 2$, and $9 \leftrightarrow -1 \leftrightarrow 3$, where the last number in each “ $\cdot \leftrightarrow \cdot \leftrightarrow \cdot$ ” comes from the set $\{1, 2, 3\} \subseteq \mathbb{N}$, such that we fit Definition 2.2.3.

Theorem 2.2.5. *The following sets are infinite: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{R}^n$ for any $n \in \mathbb{N}$.*

Remark 2.2.6. Cardinality is intuitive for finite sets, but becomes unintuitive for infinite sets. For instance, intuition tells us that $\mathbb{N}$ and $\mathbb{R}$ must have different sizes ($\mathbb{R}$ seems “denser” than $\mathbb{N}$), but how to verify that no bijection exists between the two is not obvious (with our current machinery, we would need to exhaust every candidate bijection and show none are valid, which is unfeasible). Note that it is often easier to prove equal cardinality: it suffices to construct a bijection explicitly.

Definition 2.2.7. (*Countable*). All finite sets are *countable*. An infinite set is *countably infinite* if and only if it has equal cardinality with $\mathbb{N}$. Otherwise, it is *uncountable*.

Theorem 2.2.8. $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countably infinite. $\mathbb{R}$ is uncountable. Subsets of countable sets are countable.

Remark 2.2.9. Intuitively, a countable set is one whose elements can be exhaustively enumerated through some (possibly infinite) counting process. In other words, there exists a way to exhibit two “successive” elements and associate them with two successive natural numbers (without, say, missing something in between).

Fact 2.2.10. (*Indexing*). Let S and I be sets with equal cardinality. We can then assign an element of I , called an *index*, to each element in S . We call I the *index set*. Indexing S with I is precisely constructing a bijective map $f : I \rightarrow S$ and labeling each element $x \in S$ with the associated preimage element $i = f^{-1}(x) \in I$. If S is indexed by I , we can write $S = \{x_i : i \in I\}$, or $S = \{x_i\}_{i \in I}$ in shorthand.

Remark 2.2.11. Indexing is intuitive for countable sets, as we can list out each element and assign it an index. In particular, we can always use subsets of natural numbers as the index set. Indexing is weirder for uncountable sets, as we cannot really perform the same type of explicit listing.

Remark 2.2.12. When indexing S with I , we are free to choose any bijection between them. In other words, the indexing process need not be unique or canonical.

Example 2.2.13. In statistics, datasets are often indexed by subsets of $\mathbb{N}$ starting from 1 and counting up. For instance, we may have n samples denoted $x_1, x_2, \dots, x_n$ which live in the dataset $S = \{x_i\}_{i=1}^n$. Here, $I = \{i \in \mathbb{N} : 1 \leq i \leq n\}$, and $\{x_i\}_{i=1}^n$ is common shorthand for $\{x_i\}_{i \in I}$.

Remark 2.2.14. Note that indexing a set does not impose an ordering of elements: $\{x_1, x_2, x_3\}$ is equivalent setwise to $\{x_2, x_1, x_3\}$. In other words, sets are invariant under permutations of indices, i.e. relabeling (in a sense, “visually reordering”) elements. Permuting indices only changes the specific bijection used to assign indices.

Remark 2.2.15. We can still specify rules for indexing (e.g. label integers $x_1, x_2, \dots, x_n$ such that $x_1 \leq x_2 \leq \dots \leq x_n$), but these are orderings induced by an “external” structure on the objects (e.g. their magnitudes), not by the indices themselves. Again, indexing is not an intrinsic setwise property.

2.3 “Operations” on Sets

Here I discuss what I will broadly call “operations” on sets, e.g. taking intersections, unions, differences, etc. Again, this is not precise according to a ZFC model (which establishes unions axiomatically), but to me it is somewhat helpful to modularize these as “ways to make sets from other sets” that stand apart from the Axioms of Specification and Replacement, which relate more-so to set-builder notation.

Definition 2.3.1. (*Unions*) Let A and B be sets. We call the set containing all elements of A , as well as all elements of B , the *union* of A and B , denoted $A \cup B$. Symbolically:

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

where “or” is inclusive, i.e. we include elements in both A and B . More generally, we can define the union of a collection of sets $\mathcal{A} = \{A_i\}_{i \in I}$ indexed by I as

$$\bigcup_{i \in I} A_i := \{x : x \in A_i \text{ for some } i \in I\}$$

Definition 2.3.2. (*Intersections*) Let A and B be sets. We call the set containing all elements of A , which are also elements of B , the *intersection* of A and B , denoted $A \cap B$. Symbolically:

$$A \cap B := \{x \in A : x \in B\}$$

Again, we can define the intersection of a collection of sets $\mathcal{A} = \{A_i\}_{i \in I}$ indexed by I as

$$\bigcap_{i \in I} A_i := \{x : x \in A_i \forall i \in I\}$$

Definition 2.3.3. (*Set difference*) Let A and B be sets. We call the set consisting of all elements of A that are not also in B the *set difference* of A and B , denoted $A \setminus B$. Symbolically:

$$A \setminus B := \{x \in A : x \notin B\}$$

Definition 2.3.4. Let A be a set and B be a subset of A . We call the set of all elements in A

that are not also in B the *complement* of B in A , denoted B^c or $\bar{B}$. Symbolically:

$$B^c := \{x \in A : x \notin B\} = A \setminus B$$

In particular, the complement of B in A is just the set difference of A and B specialized to the setting where B is a subset of A . Note that B^c is defined relative to a superset A , which is usually understood from context. The notation B^c makes no sense without a specified superset.

Fact 2.3.5. *Given the above definitions, let S be a set and A, B, C be subsets of S . Then the following properties hold:*

- (i) $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.
- (ii) $A \cup S = S$ and $A \cap S = A$.
- (iii) $A \cup A = A \cap A = A$.
- (iv) (Commutativity). $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
- (v) (Associativity). $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
- (vi) (Distributivity). $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- (vii) $A \cup (S \setminus A) = S$ and $A \cap (S \setminus A) = \emptyset$.
- (viii) (De Morgan). $S \setminus (A \cup B) = (S \setminus A) \cap (S \setminus B)$ and $S \setminus (A \cap B) = (S \setminus A) \cup (S \setminus B)$. In terms of complements: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

2.4 Tuples and Cartesian Products

Here I discuss the construction of tuples and Cartesian products, which are typically implicated in the first informal definitions of vectors.

Definition 2.4.1. (*Ordered pairs*). Let x and y be objects. We define an *ordered pair* as a new object denoted (x, y) , where x is its first *component* and y its second component. Two ordered pairs (x, y) and (x', y') are equal if and only if $x = x'$ and $y = y'$ (i.e. equality is defined component-wise).

Remark 2.4.2. Unlike with sets, ordering matters in ordered pairs: the first component is intrinsically distinct from the second component; (x, y) and (y, x) are not the same object in general (unless, of course, $x = y$).

Remark 2.4.3. I find it helpful to think of ordered pairs as a way of arranging information. An object with two pieces of distinctly labeled information can be represented as an ordered pair. For instance, we can write vectors in a 2-dimensional vector space as ordered pairs of coordinates with respect to a given basis. However, in general, the ordered pair itself is not the vector, and the vector is not the ordered pair; an ordered pair is just a way to write the vector.

Definition 2.4.4. (*n-tuple*). Let $n \in \mathbb{N}$ and let $\{x_i\}_{i=1}^n$ be an indexed set of objects. We defined an (ordered) *n-tuple* as a new object denoted $(x_1, \dots, x_n)$, where x_i is its *i*th component. We may abbreviate the notation to $(x_i)_{i=1}^n$. Ordered pairs $(x_i)_{i=1}^n$ and $(y_i)_{i=1}^n$ are equal if and only if $x_i = y_i$ for all $i = 1, \dots, n$.

Remark 2.4.5. Evidently, ordered pairs are a special case of *n*-tuples for $n = 2$. Again, the ordering of an *n*-tuple is an intrinsic property.

Remark 2.4.6. Again, we may think of an *n*-tuple as a way of presenting *n* pieces of information for an object. For instance, vectors in an *n*-dimensional vector space are defined by *n* coordinates with respect to a given basis, so representing them as *n*-tuples is natural. Note again that vectors themselves are not their *n*-tuple coordinate representations, which are not canonical properties as they depend on a choice of basis. Tuples are just a way of writing vectors that is highly compatible with the algebraic structure of (finite-dimensional) vector spaces.

Given these new objects, we now define the sets in which they live. Intuitively, we need to “place” two sets “side by side” for ordered pairs, as these are “composite” objects constructed from two objects placed in sequence.

Definition 2.4.7. (*Cartesian product*). Let X and Y be sets. We define the *Cartesian product* of X and Y , denoted $X \times Y$, as the set of ordered pairs (x, y) where $x \in X$ and $y \in Y$. Symbolically:

$$X \times Y := \{(x, y) : x \in X, y \in Y\}$$

Definition 2.4.8. (*n-fold Cartesian products*). Let $n \in \mathbb{N}$ and let $(X_i)_{i=1}^n$ be an *n*-tuple of sets. We define their *n-fold Cartesian product*, denoted $X_1 \times \dots \times X_n$ or $\prod_{i=1}^n X_i$, as the set of *n*-tuples $(x_i)_{i=1}^n$ where $x_i \in X_i$ for all $i = 1, \dots, n$. Symbolically:

$$\prod_{i=1}^n X_i := \{(x_i)_{i=1}^n : x_i \in X_i \text{ for all } i = 1, \dots, n\}$$

Remark 2.4.9. Note that these definitions technically assert the existence of *n*-tuples and *n*-fold Cartesian products as objects and sets respectively without proof, though both can be established from ZFC axioms.

Remark 2.4.10. An *n*-fold Cartesian product of the same set X is usually denoted $X^n := \prod_{i=1}^n X$ in shorthand.

Definition 2.4.11. $\mathbb{R}^n$ is defined to be the *n*-fold Cartesian product of *n* copies of $\mathbb{R}$, i.e. $\mathbb{R}^n := \prod_{i=1}^n \mathbb{R}$. Thus, objects in $\mathbb{R}^n$ are precisely *n*-tuples $(x_i)_{i=1}^n$ for which $x_i \in \mathbb{R}$ for all $i = 1, \dots, n$.

Remark 2.4.12. Definition 2.4.11 is why we can think of vectors in $\mathbb{R}^n$ as canonical *n*-tuples themselves, as they are precisely *n*-tuples of real numbers at an object level. This happens to coincide perfectly with their coordinate representations in the standard basis, so we often neglect

the distinction between vectors in $\mathbb{R}^n$ and their n -tuple coordinate representation. However, Remark 2.4.6 holds for abstract vector spaces, as these need not be n -fold Cartesian products of a convenient set like $\mathbb{R}$. To reiterate, in general, abstract vectors are not the same as their n -tuple coordinate representations.

Example 2.4.13. (i) The 3-fold Cartesian product $\mathbb{N} \times \mathbb{Q} \times \{\pi, e\}$ contains 3-tuples such as $(2, 0.5, \pi)$ and $(4, -0.333\dots, e)$. (ii) Let $\mathbb{Z}[x]$ denote the set of polynomials in x with integer coefficients. Then the Cartesian product $\mathbb{R} \times \mathbb{Z}[x]$ contains ordered pairs such as $(5, x^5 + 2x^2 + x)$ and $(\sqrt{2}, 7x^6 - 1)$. (iii) Let $+$ denote the addition operation between integers. Then the ordered pair $(\mathbb{Z}, +)$ may be used to denote the additive group of integers (more to come in an Abstract Algebra course).

2.5 Inheritance of Properties

Lastly, I want to discuss when subsets may inherit properties of their supersets. In Linear Algebra, this typically arises when discussing subspaces of a vector space. In particular, to show that a subset of a vector space is a subspace, we only need to check certain properties, while others are *inherited*.

Inheritance is never purely set theoretic; sets have too little structure to inherit. We need only worry about when sets endowed with certain properties may pass certain properties on to the subsets “automatically.” Below are some informal, “works most of the time” rules, which suffices for our Linear Algebra context:

- Properties which hold for all elements of a set S , and which do not pertain to membership in S , e.g. associativity of addition in vector spaces, are inherited by subsets of S . Intuitively, all elements of $A \subseteq S$ are also elements of S , so as long as we are certain the “for all” property does not mess with membership in A , we can typically conclude that all elements of A also satisfy the property.
- Properties which hold for some element in a set S , i.e. “there exists” statements, are not inherited by subsets of S , as these may not contain the distinguished elements in S which satisfy the property.
- Properties pertaining to membership in a set S , e.g. “ S is closed under addition,” are not inherited by subsets of S , as membership in S does not guarantee membership in subsets of S .

Example 2.5.1. Let V be a vector space and $W \subseteq V$ be a subset of V . (i) The associativity, distributivity, and commutativity of vector addition is inherited from V , as these pertain to all elements of V and do not mess with membership. (ii) The closure of W under vector addition and scalar multiplication is not inherited from V , as these pertain to membership in V . (iii) The existence of an additive identity in W is not inherited from V , as this is a classic “there exists” statement. (iv) The existence of additive inverses for all vectors in W is not inherited from V , as this is still a “there exists” statement.

Example 2.5.1 is precisely why we must check the existence of the additive identity, plus closure under vector addition and scalar multiplication, to prove that a subset of a vector space is a subspace.

Note that the existence of additive inverses is covered by closure under scalar multiplication, as multiplying any vector by -1 produces its additive inverse (this is really about the properties of the additive inverse of the multiplicative identity in the underlying field of a vector space, which always exists). Thus checking $cx \in W$ for all $x \in W$ and scalars c suffices, as we simply specialize c to -1 to exhibit additive inverses.

3 Proofs with Sets

Here I will discuss some proof structures typically encountered in introductory Linear Algebra involving sets, and walk through some examples. Most involve steps validated by the foundational set theory from Sections 1 and 2, despite not directly requiring knowledge of this theory to perform.

3.1 Set Equality

Consider the following problem structure:

Problem 3.1.1. Let S be a set, and let $X := \{x \in S : P(x)\}$ and $Y := \{y \in S : Q(y)\}$, where $P(x)$ and $Q(y)$ are properties of x and y respectively. Prove that $X = Y$.

By Axiom 1.2, we need to show that every $x \in X$ is also in Y , and every $y \in Y$ is also in X . In practice, we use a trivial proposition on subset containment.

Proposition 3.1.2. *Let X and Y be sets. If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$.*

Proof. If $x \in X$, then since $X \subseteq Y$, we have $x \in Y$ by definition of subsets. If $y \in Y$, then since $Y \subseteq X$, we have $y \in X$ by definition of subsets. Thus $X = Y$. $\square$

Thus, to construct a proof of Problem 3.1.1 (for the case where X and Y are nonempty; if both are empty, the argument is trivial), we follow these general steps:

- 1a. Invoke Single Choice to take an arbitrary $x \in X$.
- 1b. Use the fact that $P(x)$ holds to argue that $Q(x)$ holds.
- 1c. $Q(x)$ holding implies $x \in Y$.
- 2a. Invoke Single Choice to take an arbitrary $y \in Y$.
- 2b. Use the fact that $Q(y)$ holds to argue that $P(y)$ holds.
- 2c. $P(y)$ holding implies $y \in X$.

Example 3.1.3. Let $v \in \mathbb{R}^n$ and $c \in \mathbb{R}$, where $v \neq \mathbf{0}$ and $c \neq 0$. Let $X := \text{span}(v)$ and $Y := \text{span}(cv)$. Prove that $X = Y$.

Proof. Recall that by definition, $X = \{av : a \in \mathbb{R}\}$ and $Y = \{b(cv) : b \in \mathbb{R}\}$. Take $x \in X$. If $x = \mathbf{0}$, then $x = 0(cv)$, so $x \in Y$. If $x \neq \mathbf{0}$, then $x = av$ for some nonzero $a \in \mathbb{R}$. Since a and c are both nonzero, $a/c \in \mathbb{R}$ is well-defined, so we can write $x = (a/c)(cv)$, so $x \in Y$ (namely, $x = b(cv)$ for $b = a/c \in \mathbb{R}$). Thus $X \subseteq Y$.

Now take $y \in Y$. If $y = \mathbf{0}$, then $y = 0(v)$, so $y \in X$. If $y \neq \mathbf{0}$, then $y = b(cv) = (bc)v$ for some nonzero $b \in \mathbb{R}$, so $y \in X$ (namely, $y = av$ for $a = bc \in \mathbb{R}$). Thus $Y \subseteq X$. Hence by Proposition 3.1.2, we conclude $X = Y$. $\square$

Example 3.1.4. Let A be an $m \times n$ real matrix, and let $N(A) := \{x \in \mathbb{R}^n : Ax = 0\}$ be the null space of A . Prove that $N(A) = N(A^T A)$.

Proof. Take $x \in N(A)$, then $Ax = \mathbf{0} \implies A^T Ax = A^T \mathbf{0} = \mathbf{0}$, so $x \in N(A^T A)$ and $N(A) \subseteq N(A^T A)$. Now take $x \in N(A^T A)$, then $A^T Ax = \mathbf{0}$, so we have

$$x^T(A^T Ax) = x^T \mathbf{0} = \mathbf{0} \implies (Ax)^T(Ax) = \|Ax\|^2 = \mathbf{0}$$

But $\|y\|^2 = y \cdot y = \mathbf{0} \iff y = \mathbf{0}$ for any $y \in \mathbb{R}^m$ by the positive-semidefinite property of dot (inner) products, meaning $Ax = \mathbf{0}$, so $x \in N(A)$ and thus $N(A^T A) \subseteq N(A)$. Now by Proposition 3.1.2 we know $N(A) = N(A^T A)$. $\square$

Remark 3.1.5. A consequence of $N(A) = N(A^T A)$ is that $\text{rank}(A) = \text{rank}(A^T A)$ by the Rank-Nullity Theorem, meaning $A^T A$ is invertible if and only if A has full rank. The matrix $A^T A$ is called a *Gram* matrix, and appears frequently in statistics. In particular, $(A^T A)^{-1}$ is implicated in the *ordinary least squares* (OLS) solution to the standard linear regression problem, where the columns of A represent the observed predictor variables. Hence the OLS solution exists (and is unique) if and only if the observed predictors are linearly independent vectors (i.e. we have non-perfect collinearity).

3.2 Is a Subset a Subspace?

Consider the following problem structure:

Problem 3.2.1. Let V be a vector space, and let $X := \{x \in V : P(x)\}$, where $P(x)$ is a property of x . Prove that X is a subspace of V .

As seen in Example 2.5.1, since X is a subset of V , a vector space (which we take to be real, i.e. over $\mathbb{R}$, but can be over any field), we only need to check the following:

1. Additive identity (and nonempty): The $\mathbf{0}$ vector is in X , i.e. $P(\mathbf{0})$ holds.
2. Closure under vector addition: Invoke Single Choice to take arbitrary $x, y \in X$. Then $x + y \in X$.
3. Closure under scalar multiplication: Invoke Single Choice to take an arbitrary $x \in X$ and $c \in \mathbb{R}$. Then $cx \in X$.

Remark 3.2.2. We may equivalently combine the check of closure under vector addition and closure under scalar multiplication by taking arbitrary $x, y \in X$ and $c \in \mathbb{R}$, and showing $x + cy \in X$. Note that this suffices for both closures, as taking $c = 1$ shows closure under vector addition, and taking $x = \mathbf{0}$ gives closure under scalar multiplication.

Example 3.2.3. Let $V = C^1(\mathbb{R})$, the space of all continuously differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, which gives a vector space over $\mathbb{R}$. For a fixed $\alpha \in \mathbb{R}$, let $X := \{f \in C^1(\mathbb{R}) : f'(\alpha) = 0\}$, where f' denotes the first derivative of f . Prove that X is a subspace.

Proof. Take the zero function $0(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ given by $0(x) = 0$ for all $x \in \mathbb{R}$. This is evidently the additive identity, and $0'(x) = 0$ for all $x \in \mathbb{R}$, so $0(\cdot) \in X$. Now take $f, g \in X$ and $c \in \mathbb{R}$. Then, by the linearity of differentiation, we know $(f + cg)' = f' + cg'$, so $(f + cg)'(\alpha) = f'(\alpha) + cg'(\alpha) = 0 + c0 = 0$. Hence $f + cg \in X$, so X is indeed closed under vector addition and scalar multiplication. Hence we conclude X is a subspace of $C^1(\mathbb{R})$. $\square$

3.3 Disproving Structural Properties

Consider the following problem structure:

Problem 3.3.1. Let V be a vector space, and let $X := \{x \in V : P(x)\}$, where $P(x)$ is a property of x . Show that X is NOT a subspace of V .

Remark 3.3.2. Often times, a problem like Problem 3.3.1 is posed as a “prove or give a counterexample” problem.

As seen in the discussion following Problem 3.2.1, we just need to contradict one of three non-inherited properties of subspaces. It suffices to identify a single counterexample where one of the properties fails to hold, so the cleanest proof method is to invoke Distinguished Choice to pick specific elements. In some trivial cases, it is easy to show that the additive identity vector is missing.

Example 3.3.3. Let $V = \mathbb{R}^2$ and let $X := \{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$. Show that X is not a subspace of $\mathbb{R}^2$.

Proof. Take $v = (-1, 0)$ and $w = (0, 1)$, where since $-1 \cdot 0 = 0 \cdot 1 = 0 \geq 0$, we know $v, w \in X$. However, $v + w = (-1, 1)$, and $-1 \cdot 1 = -1 \not\geq 0$, so $v + w \notin X$. Thus X is not closed under vector addition, meaning X is not a subspace of $\mathbb{R}^2$. $\square$

Remark 3.3.4. Observe that in Example 3.3.3, both the additive identity and closure under scalar multiplication holds for X , so counterexamples only exist for closure under vector addition.

Consider the following problem structure:

Problem 3.3.5. Let V and W be subspaces, and let $T : V \rightarrow W$ be a function given by $x \mapsto T(x)$ for all $x \in V$. Is T a linear transformation?

Recall that T is called a linear transformation if $T(x + y) = T(x) + T(y)$ and $T(cx) = cT(x)$ for all $x, y \in V$ and scalars c . Similar to the reduction in Remark 3.2.2, we may combine these criterion when checking linearity, i.e. if $T(x + cy) = T(x) + cT(y)$ for all $x, y \in V$ and scalars c then T is linear. Again, taking $c = 1$ and $x = \mathbf{0}$ produces the two linearity conditions respectively. Similarly, if T is not linear, it suffices to identify one specific combination of x, y and c such that $T(x + cy) \neq T(x) + cT(y)$.

Example 3.3.6. Let $V = C^1[-1, 1]$, the set of continuous real-valued functions on the interval $[-1, 1] \subset \mathbb{R}$. For all $f \in V$, define $T : V \rightarrow V$ by

$$T(f)(x) = \frac{f(x) + f(-x)}{2}, \quad x \in [-1, 1]$$

Is T a linear transformation?

Proof. T is linear. Take $f, g \in V$ and $c \in \mathbb{R}$. We check that

$$\begin{aligned} T(f + cg)(x) &= \frac{(f + cg)(x) + (f + cg)(-x)}{2} \\ &= \frac{(f(x) + cg(x)) + (f(-x) + cg(-x))}{2} \\ &= \frac{f(x) + f(-x)}{2} + c \frac{g(x) + g(-x)}{2} = T(f)(x) + cT(g)(x) \end{aligned}$$

□

Example 3.3.7. Let $V = \mathbb{R}^2$. For all $x = (x_1, x_2) \in \mathbb{R}^2$, define $T_1, T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T_1(x) = (x_1 + 2x_2, x_2 - x_1), \quad T_2(x) = (x_1 + 2x_2, x_2 - x_1 + 1)$$

Is T_1 a linear transformation? Is T_2 a linear transformation?

Proof. T_1 is linear. Take $x, y \in \mathbb{R}^2$ and $c \in \mathbb{R}$. We check that

$$T_1(x + cy) = \begin{bmatrix} (x_1 + cy_1) + 2(x_2 + cy_2) \\ (x_2 + cy_2) - (x_1 + cy_1) \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ x_2 - x_1 \end{bmatrix} + c \begin{bmatrix} y_1 + 2y_2 \\ y_2 - y_1 \end{bmatrix} = T_1(x) + cT_1(y)$$

However, T_2 is NOT linear. Take $\mathbf{0} \in \mathbb{R}^2$. Check that

$$T_2(\mathbf{0} + \mathbf{0}) = (0, 1), \quad T_2(\mathbf{0}) + T_2(\mathbf{0}) = (0, 2) \neq T_2(\mathbf{0} + \mathbf{0})$$

□

3.4 Injectivity, Surjectivity, Bijectivity

Consider the following problem structure:

Problem 3.4.1. Let $f : S \rightarrow S'$ be a function. Is f injective? Surjective? Bijective?

Again, we present a couple useful theorems below. The first is a tautology and itself a great exercises in manipulating the key definitions, as well as effectively choosing elements. I provide a tedious version of the proof. The second involves more analysis machinery to prove elegantly; there are many proofs online for reference.

Theorem 3.4.2. Let $f : S \rightarrow S'$ be a function. Then f is bijective if and only if it is invertible.

Proof. First suppose f is bijective. Let $g : S' \rightarrow S$ be given by $x' \mapsto x$ such that $x' = f(x)$ for all $x' \in S'$. Since f is surjective, we know there exists $x \in S$ such that $f(x) = x'$ for all $x' \in S'$. Since f is injective, we also know $f(x) = f(y) \implies x = y$, meaning the element $x \in S$ such that $f(x) = x'$ is also unique for all $x' \in S'$. Hence g is a well-defined function.

Now we show $g = f^{-1}$ the inverse of f . Take $x \in S$ and let $x' = f(x)$. Then $(g \circ f)(x) = g(f(x)) = g(x') = x$ by definition of g , so $g \circ f = \text{id}_S$. Then take $x' \in S'$ and let $x = g(x')$. Then $(f \circ g)(x') = f(g(x')) = f(x) = x'$ by definition of g again, so $f \circ g = \text{id}_{S'}$. Thus by Definition 2.1.14 we know $g = f^{-1}$. In particular, f has an inverse, so f is invertible.

Now, suppose f is invertible, i.e. there exists $f^{-1} : S' \rightarrow S$ such that $f^{-1} \circ f = \text{id}_S$ and $f \circ f^{-1} = \text{id}_{S'}$. To show f is surjective, take $x' \in S'$ and let $x = f^{-1}(x')$. Then $f(x) = f(f^{-1}(x')) = x'$, meaning for any $x' \in S'$, there exists $x \in S$ (namely $f^{-1}(x')$) such that $x' = f(x)$, so f is surjective by Definition 2.1.10.

To show f is injective, suppose $x, y \in S$ such that $f(x) = f(y)$. Then applying f^{-1} to both sides gives $f^{-1}(f(x)) = f^{-1}(f(y)) \implies x = y$. Thus f is injective by Definition 2.1.9. Hence f is bijective by Definition 2.1.11. We have now show both directions, and conclude the proof. $\square$

Theorem 3.4.3. *Let S and S' be finite sets of equal cardinality. Then a function $f : S \rightarrow S'$ is injective if and only if it is surjective.*

Proof. Find online. $\square$

We may use Theorem 3.4.2 and 3.4.3 in some contexts to simplify proofs. For instance, if $f : S \rightarrow S'$ where $|S| = |S'|$, then by Theorem 3.4.3, we only need to show f is either injective or surjective to conclude f is bijective. Likewise, Theorem 3.4.2 implies that to prove f is bijective, it suffices to exhibit its inverse.

Example 3.4.4. Let A be a real invertible $n \times n$ matrix. Define $T : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ by

$$T(X) = AXA^{-1}$$

for all $X \in \mathbb{R}^{n \times n}$. Note T is linear by the linearity of matrix multiplication. Prove that T is bijective.

Proof. Notice that the map $H : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ given by $H(X) = A^{-1}XA$ satisfies

$$H(T(X)) = A^{-1}AXA^{-1}A = X, \quad T(H(X)) = AA^{-1}XAA^{-1} = X$$

so $H = T^{-1}$. In particular, T is invertible, so by Theorem 3.4.2, T is bijective. $\square$

Example 3.4.5. (An illustrative number theory problem). Let $\mathbb{Z}_n$ denote the set of residues modulo n . In human language, for $n \in \mathbb{N}$,

$$\mathbb{Z}_n = \{p \in \mathbb{Z} : 0 \leq p \leq n-1\} \subset \mathbb{Z}$$

Note that equality in $\mathbb{Z}_n$ is established modulo n . Two integers p, q are said to be *congruent modulo n* if they yield equal remainders after division by n . For this, we write $p \equiv q \pmod{n}$. Lastly, for integers p and q , we write $p|q$ to mean p divides q with no remainder. In particular, $q \equiv 0 \pmod{p}$ if and only if $p|q$.

Now let M, N be distinct positive integers. Suppose M, N are relatively prime, meaning their greatest common divisor is 1. Prove that every $x \in \mathbb{Z}_{MN}$ can be written uniquely in the form

$$x = aM + bN \quad \text{for some } 0 \leq a \leq N-1, 0 \leq b \leq M-1$$

Proof. Given M, N positive, distinct, relatively prime integers, consider the set

$$S := \{(a, b) : 0 \leq a \leq N-1, 0 \leq b \leq M-1\} = \mathbb{Z}_M \times \mathbb{Z}_N$$

Define a map $\varphi : S \rightarrow \mathbb{Z}_{MN}$ by

$$\varphi(a, b) = aM + bN \pmod{MN}$$

We prove that φ is injective. Suppose $a_1M + b_1N \equiv a_2M + b_2N \pmod{MN}$. Then $q := (a_1 - a_2)M + (b_1 - b_2)N \equiv 0 \pmod{MN}$, so $MN|q$, meaning $M|q$ and $N|q$. Since $M|q$,

$$(a_1 - a_2)M + (b_1 - b_2)N \equiv 0 \pmod{M} \implies (b_1 - b_2)N \equiv 0 \pmod{M}$$

Thus either $M|(b_1 - b_2)$ or $M|N$, but M and N are relatively prime, so we need $M|(b_1 - b_2) \implies b_1 \equiv b_2 \pmod{M}$. By a symmetric argument, since $N|q$, we have $a_1 \equiv a_2 \pmod{N}$. Thus φ is injective.

Now observe that $|S| = |\mathbb{Z}_M \times \mathbb{Z}_N| = MN = |\mathbb{Z}_{MN}|$, so by Theorem 3.4.3, φ is also surjective. This concludes the proof. In particular, surjectivity implies any $x \in \mathbb{Z}_{MN}$ can be written as $x = \varphi(a, b) = aM + bN$ for some $(a, b) \in S$, while injectivity implies that such an expression is unique. $\square$

3.5 Images and Kernels

Consider the following problem structure:

Problem 3.5.1. Let V and W be subspaces, and let $T : V \rightarrow W$ be a linear transformation. Identify the image and kernel of T .

Recall that since images and kernels are sets, we must use Proposition 3.1.2, i.e. prove containment both ways. Note that if showing T is injective is easier, we may take advantage of the proposition below.

Proposition 3.5.2. Let V and W be vector spaces, and let $T : V \rightarrow W$ be a linear transformation. Then $\ker T = \{0\}$ if and only if T is injective.

Proof. Suppose $\ker T = \{\mathbf{0}\}$, and suppose $T(x) = T(y)$ for some $x, y \in V$. Then $T(x - y) = \mathbf{0}$ by linearity, so $x - y \in \ker T \implies x - y = \mathbf{0} \implies x = y$, meaning T is injective.

Now suppose T is injective and take $x \in \ker T$, so $T(x) = \mathbf{0}$. Note that $T(\mathbf{0}) = \mathbf{0}$ (as $T(\mathbf{0}) = T(x - x) = T(x) - T(x) = \mathbf{0}$ for any $x \in V$). Hence $T(x) = \mathbf{0} \implies x = \mathbf{0} \implies \ker T \subseteq \{\mathbf{0}\}$. Now $T(\mathbf{0}) = \mathbf{0}$ also implies $\{\mathbf{0}\} \subseteq \ker T$, so indeed $\ker T = \{\mathbf{0}\}$. $\square$

Example 3.5.3. Let $\mathcal{P}_2$ denote the set of polynomials with real coefficients of degree at most 2. Check that $\mathcal{P}_2$ is a vector space over $\mathbb{R}$ spanned by $\{1, t, t^2\}$. Define $T : \mathcal{P}_2 \rightarrow \mathbb{R}$ by

$$T(p) = \int_0^1 p(x) dx$$

It is easy to check T is linear by the linearity of integration. Identify the image and kernel of T . In particular, exhibit a basis for each.

Proof. We see that image $T = \mathbb{R}$, as for any $c \in \mathbb{R}$, taking the constant polynomial $p_c(x) = c \in \mathcal{P}_2$ gives

$$T(p_c) = \int_0^1 c dx = c$$

meaning for all $c \in \mathbb{R}$, there exists $p_c \in \mathcal{P}$ such that $T(p_c) = c$. As a real vector space, a basis for $\mathbb{R}$ is simply $\{1\}$. The kernel, by definition, is given by

$$\ker T = \left\{ p \in \mathcal{P}_2 : \int_0^1 p(x) dx = 0 \right\}$$

To identify a basis, notice that if we pass an arbitrary $p(x) = a + bx + cx^2 \in \mathcal{P}_2$ through T , we obtain

$$T(p) = \int_0^1 a + bx + cx^2 dx = a + \frac{1}{2}b + \frac{1}{3}c$$

Thus for $T(p) = 0$, we need

$$a = -\frac{1}{2}b - \frac{1}{3}c \implies p(x) = b \left(t - \frac{1}{2} \right) + c \left(t^2 - \frac{1}{3} \right)$$

Thus, we claim that $\ker T = \text{span}(B)$ for

$$B = \left\{ t - \frac{1}{2}, t^2 - \frac{1}{3} \right\}$$

i.e. B forms a basis for $\ker T$. It is easy to check B is linearly independent. We have already shown $p \in \ker T \implies p \in \text{span}(B)$. It is straight forward to check $q \in \text{span}(B) \implies T(q) = 0$ as well, meaning indeed $\ker T = \text{span}(B)$. In particular, notice $\dim \ker T = 2$, meaning $\ker T$ forms a plane in the 3-dimensional $\mathcal{P}_2$ space. $\square$

4 Exercises

I leave a few exercises here. Some involve belaboring set theoretic tautologies. Some are more intricate algebra problems and may seem odd at a glance, but remain very doable given a solid understanding of linear algebra definitions alongside what we discussed previously. I hope solving these can provide some confidence. Solutions can be found for most of these problems online. LLMs should be able to provide solid solutions as well.

4.1 Some Tautologies

Problem 4.1.1. Let S be a set and let $A \subseteq S$. Prove that $A \cup (S \setminus A) = S$. Prove that $A \cap (S \setminus A) = \emptyset$ (from Fact 2.3.5).

Problem 4.1.2. Let A, B, C be sets. Show that $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

Problem 4.1.3. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. Prove that $\text{image}(g \circ f) = g(\text{image}(f))$ (from Fact 2.1.20).

Problem 4.1.4. Let S be a set and let $A, B \subseteq S$. Let $f : S' \rightarrow S$ be a function. Prove that $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$. Prove that $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$. Note both these sets are subsets of S' , the domain of f .

Problem 4.1.5. Prove that $f : S \rightarrow S'$ is injective if and only if $f \circ g_1 = f \circ g_2 \implies g_1 = g_2$ for all functions $g_1, g_2 : A \rightarrow S$ for any set A .

Problem 4.1.6. Prove that $f : S \rightarrow S'$ is surjective if and only if $h_1 \circ f = h_2 \circ f \implies h_1 = h_2$ for all functions $h_1, h_2 : S' \rightarrow B$ for any set B .

4.2 Some More Interesting Problems

Problem 4.2.1. Let V, W, U be vector spaces and let $T_1 : V \rightarrow W$ and $T_2 : V \rightarrow U$ be linear transformations. Define $T : V \rightarrow W \times U$ by $x \mapsto (T_1(x), T_2(x))$. Prove that $\ker T_1 \cap \ker T_2 = \ker T$.

Problem 4.2.2. Let V be a real vector space. Define

$$S := \{x \in V : f(x) = 0 \text{ for all linear maps } f : V \rightarrow \mathbb{R}\}$$

Show that $S = \{0\}$.

Problem 4.2.3. Let $\{v_1, \dots, v_k\}$ be a linearly dependent set of vectors in $\mathbb{R}^n$. Prove that for some $1 \leq i \leq k$, we can write $v_i = Ax$ for some $x \in \mathbb{R}^n$ and some $n \times n$ real matrix A where $v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_k$ are all in its column space.

Problem 4.2.4. Let $\mathbb{R}[x]$ denote the set of all polynomials in x with real coefficients. For some $a \in \mathbb{R}$, define the *substitution map* $\rho_a : \mathbb{R}[x] \rightarrow \mathbb{R}$ given by $p(x) \mapsto p(a)$. In other words, given

a degree n polynomial $p(x) = c_n x^n + \cdots c_1 x + c_0 \in \mathbb{R}[x]$, $\rho_a(p)$ evaluates $p(x)$ at $x = a$. Prove that ρ_a is a linear map (take it to be true that $\mathbb{R}[x]$ is a vector space over $\mathbb{R}$). Is ρ_a injective? Surjective? Determine $\ker \rho_a$.

As an extension, let $a_1, \dots, a_k \in \mathbb{R}$. Define the function $\rho : \mathbb{R}[x] \rightarrow \mathbb{R}^k$ by

$$p(x) \mapsto (p(a_1), \dots, p(a_k))^T$$

Prove that ρ is also a linear map (*Hint*. Just do the same thing k times).

Problem 4.2.5. Let $(V_i)_{i=1}^n$ be an n -tuple of vector spaces over $\mathbb{R}$. Prove that $\prod_{i=1}^n V_i$ is a vector space. If $V_1, \dots, V_n$ are finite-dimensional, prove that

$$\dim \left(\prod_{i=1}^n V_i \right) = \sum_{i=1}^n \dim V_i$$

(*Hint*. For the first part, recall the vector space axioms. You must define what vector addition and scalar multiplication means in the n -fold Cartesian product. For the second part, recall that $\dim V_i$ is the cardinality of any basis for V_i if V_i is finite-dimensional).

Problem 4.2.6. Let V and W be real vector spaces with bases $\mathcal{V} := \{v_1, \dots, v_n\}$ and $\mathcal{W} := \{w_1, \dots, w_m\}$ respectively. Let $T : V \rightarrow W$ be a linear transformation. Prove that

$$x := \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \ker T \iff x_1 v_1 + \cdots + x_n v_n \in \ker T$$

(*Hint*. For the “ $\Leftarrow$ ” direction, use the $m \times n$ real matrix A that represents T with respect to the bases $\mathcal{V}$ and $\mathcal{W}$).

Problem 4.2.7. Let V and W be subspaces of $\mathbb{R}^n$. Recall that we define the addition of subspaces by $V + W := \{v + w : v \in V, w \in W\} \subseteq \mathbb{R}^n$. Prove that $\dim(V + W) = \dim V + \dim W - \dim(V \cap W)$. (*Hint*. Use the bases of V and W and apply the Extension of Basis Theorem to construct a basis for $V + W$).

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